

15+ Years of Applied AI — From Machine Vision to Autonomous Geospatial Systems

repo: github.com/kamangir

Experience

2025–current **Deputy CEO, Karband Eng. Co., Alborz, Iran.**

I lead initiatives in Geospatial AI and Unmanned Ground Vehicles (UGVs) — developing intelligent systems that move, sense, and learn from the world around them. My work bridges robotics, spatial data, and applied machine learning to create technologies that operate seamlessly in real environments. Alongside this, I'm passionate about AI education for youth. I believe that teaching kids to think in systems — to design, build, and experiment — is key to nurturing a generation capable of shaping technology with creativity and purpose. At the heart of my work is a simple idea: technology should serve both intelligence and integrity. Whether through autonomous robots, spatial analytics, or educational projects, I aim to help build an ecosystem of AI that is local, sustainable, and deeply human in its intention.

2022–2025 **Staff Software Engineer, EarthDaily Analytics, Vancouver, Canada.**

EarthDaily observes, verifies, & predicts changes to the Earth's surface to help people understand & take action.

2022 **Senior Machine Learning Engineer (Computer Vision), Vivid Machines, Toronto, Canada.**

Smart technology to help fruit and vegetable farmers optimize quality and yield.

2020–2022 **Vice President, Data Science, Savormetrics Inc., Mississauga, Canada.**

Deep Learning + Food Inspection = Increasing Profits. Reducing Waste. Improving Customer Satisfaction.

2019–2020 **Lead Data Scientist, Betterview Marketplace, San Diego, USA.**

Design, development, and deployment of a deep learning framework that consumes aerial images and other types of data relevant to the insurance industry and produces property insights.

2016–2019 **Senior Scientific Developer, Fio Corporation, Toronto, Canada.**

Design, development, and deployment of a learning-enabled machine vision system that visually identifies and analyzes rapid diagnostics tests (rdt) for infectious diseases such as HIV, Malaria, Dengue, Zika, and others. At inference time, the ML models are run on an android device with limited network and power access. Therefore, battery usage optimization and tolerance to long periods of disconnectivity were essential considerations. Traceability management for data and models was a vital component of this work due to national and international regulatory requirements.

- 2015–2016 **Research Scientist**, *Intellijoint Surgical, Waterloo, Canada.*
Extensions of the capabilities of a monocular infrared tracking system that was a component within the surgical navigation product that utilized machine vision to carry out and confirm geometrical measurements in-vivo. Also, worked on the inertial data that was produced by the system for validation and augmentation. Parameter estimation using least mean square and Levenberg-Marquardt cost function minimization.
- 2009–2015 **Researcher**, *Imaging Group, Epson Canada Limited, Toronto, Canada.*
Research and development on future products and concepts in the fields of visual inspection, symbology detection, 3d object detection and pose estimation, camera calibration (monocular, stereo, and depth/range), augmented and virtual reality, 3d scan/display systems (stereo, time-of-flight, structured light projection, and other depth sensors), depth processing (including bilateral upsampling, filtering, and registration and fusion of multiple depth and flat cameras using different variations of iterative closest point - icp), head-mounted displays, and robotics. Stochastic optimization in the presence of outliers using RANSAC and robustified cost minimization.
- 2001–2004 **Process Control Engineer**, *Karband Eng. Co., Alborz, Iran.*
Design, implementation and erection of PLC-based control systems for medium-sized machinery in pipe and profile production plants.
- 1998–2001, 2004–2009 **Research Assistant**, *University of Manitoba, Telecommunications Research Laboratories (TRLabs) Winnipeg and Biomechanics Laboratory, Sharif University of Technology.*
Network optimization, earthquake damage detection using satellite/aerial imagery, human gait analysis, 3d surface reconstruction, color image watermarking and data-hiding, visual encryption, image compression, color transfer, grayscale image colorization, computational photography, fuzzy clustering, skin detection, and pornography classification. Fuzzy modelling of multi-layer systems, especially within the field of pattern recognition, using Bayesian models.

Technical Skills

Ummanned Ground Vehicles (UGV), *differential drive, multi-threaded vision and ultrasonic sensors, pytorch models, localized positioning systems, power management, field tests.*

Deep Learning, *object detection, object recognition, and semantic segmentation on satellite and aerial, mobile and static, RGB and multi-spectral cameras in production for insurance and agg.*

GeoSpatial AI, *active deep learning experience on Venus, Maxar (WorldView), Sentinel-1 and Sentinel-2, RCM, multi-spectral optical and SAR, in Python and Bash on the cloud for semantic segmentation, change detection, carbon sequestration, object detection, surveillance, and tasking.*

Geospatial, *rasterio, geojson, shape files, xarray, QGIS, gdal.*

Insurance, *aerial image analysis for roof and yard content and damage assessment and object detection, risk modelling.*

Agritech, *vertical farm/orchard vision applications, tree detection and localization, seed/blossom/fruit counting, size estimation, and plant statistics.*

Machine Vision, *camera models, triangulation, pose estimation.*

Machine Learning, *TensorFlow, PyTorch, MLFlow, Fiftyone, Roboflow, Yolo.*

Python, *OpenCV, Boto3, PyQt, PyMySQL, NumPy, SciPy, Matplotlib, scikit-learn, Pandas, Seaborn, Jupyter Notebooks, Django.*

Linux, *bash programming, Ubuntu, Raspbian.*

Cloud, *Amazon S3, Amazon SQS, Amazon EC2, GCS, slurm, Amazon SageMaker, AWS Batch, Argo Workflows.*

Databases, *MySQL, Amazon RDS, PostgreSQL.*

Software Development, *agile development, JIRA, git, gitlab, papertrail, ci/cd, docker.*

Hardware, *Jetson, integration w/ cameras and sensors, Raspberry Pi, 3D design and print, general electronics, general digital electronics, general circuits.*

Web, *Django, Flask, php.*

Other, *L^AT_EX, POV-Ray, IVT, PCL, Clmg, OpenGL, Armadillo, Gandalf, Ceres Solver, and other open-source tools.*

Education

- 2005–2009 **Ph.D.**, *The University of Manitoba, Canada.*
Electrical and Computer Engineering Department
- 2003–2005 **Master of Science**, *Sharif University of Technology, Iran.*
Mathematics Science Department, Computer Science Group (Scientific Computation)
- 1996–2003 **Bachelor of Science**, *Sharif University of Technology, Iran.*
Electrical Engineering Department, Control Group

Certifications

- 2022 **AWS Cloud Technical Essentials.**
By Amazon Web Services on Coursera
- 2019 **Deep Learning Specialization, Five-Course Specialization.**
By Andrew Ng (deeplearning.ai)
- 2017 **Neural Networks for Machine Learning.**
By the University of Toronto on Coursera
- 2017 **Machine Learning.**
By Stanford University on Coursera
- 2017 **Machine Learning, Four-Course Specialization.**
By the University of Washington on Coursera

Patents

"System, method and/or computer readable medium for non-invasive workflow augmentation", WO Application Number WO2018094534A1, Priority Date 26 November 2016.

"System, method and/or computer-readable medium for identifying and/or localizing one or more rapid diagnostic tests", WO Application Number WO2018094533A1, Priority Date 26 November 2016.

"Visual pattern recognition system, method and/or computer-readable medium", WO Application Number WO2018094532A1, Priority Date 26 November 2016.

"Systems and methods for tracker characterization and verification", US Application Number US20170345177A1, Priority Date 27 May 2016.

"Systems, methods and devices to scan 3d surfaces for intra-operative localization", International Publication Number WO2017185170A1, Priority Date 28 April 2016.

"Method for object pose estimation, apparatus for object pose estimation, method for object estimation pose refinement and computer readable medium", Japanese Patent JP2013050947A, Publication Date 19 October 2016.

"HMD Calibration with Direct Geometric Modeling", US Patent US20160012643A1, Publication Date 14 January 2016.

"HMD Calibration with Direct Geometric Modeling", EU Patent No. 15175799.4 - 1902, Filing Date 8 July 2015.

"System generating three-dimensional model, method and program", Japanese Patent JP2015176600A, Publication Date 5 October 2015.

"Holocam Systems and Methods", US Patent US20150261184, Publication Date 17 September 2015.

"Method and Apparatus for Improved Training of Object Detecting System", US Patent US20140079314, Publication Date 20 March 2014.

"Method for simulating impact printer output, evaluating print quality, and creating teaching print samples", US Patent 8654398, Publication Date 18 February 2014.

"Method and apparatus for object pose estimation", US Patent 8467596, Publication Date 18 June 2013.

Select Publications

Conference Poster, *Quantitative Measurement Of The Impact Of Five Degradations On Deep Learning Detection Of Aircraft In Medium-Resolution Satellite Imagery*, Arash Abadpour and Chris Rampersad, International Geoscience and Remote Sensing Symposium, Pasadena, California, 2023 (IGARSS 2023).

Journal Paper, *On Applications of Pyramid Doubly Joint Bilateral Filtering in Dense Disparity Propagation*, Arash Abadpour, 3D Research, Volume 5, Issue 2, 25 April 2014, Pages 1–20.

Ph.D. Thesis, *QoS-Constrained Information Theoretic Capacity Maximization in CDMA Systems*, Electrical and Computer Engineering Department, University of Manitoba, Winnipeg, Manitoba, Canada, Supervised by Prof. Attahiru Sule Alfa (Ph.D.), 2005–2009.

M.Sc. Thesis, *Color Image Processing using Principal Component Analysis*, Mathematics Science Department, Sharif University of Technology, Tehran, Iran, Supervised by Shohreh Kasaei (Ph.D.) and A. Daneshgar (Ph.D.), 2004–2005.

more: Google Scholar

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